Formula 1 Podium Prediction Companion

Milestone 1 — Analytical report

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Task: predict whether a driver finishes on the podium (`positionOrder ≤ 3`) using only information available after qualifying and before the race starts. Unit of analysis: one driver in one race.

1. Data and cleaning audit

The dataset is the Ergast Formula 1 World Championship data (1950–2024): 14 linked tables, 1,125 races, 861 drivers, 212 constructors and 26,759 result rows. Raw CSVs are read as plain text into an immutable snapshot, and every cleaning step is applied to copies.

1.1 Sentinels and semantic typing

The files use the literal string `\N` for “no value”, which pandas does not treat as missing. 30 of 120 columns contain it, 160,076 cells in total. The worst columns are not data errors but structural: `constructor_results.status` (99.9%), sprint and practice date/time columns in `races` (92–99%, because these sessions only exist in recent seasons), `drivers.number` (93%), and `results.time` / `milliseconds` (71%, because only the lead lap finishes have a race time). Missingness also depends on the era (figure 2), so it is informative and we do not impute at the cleaning stage: a missing value means “does not exist”, not zero.

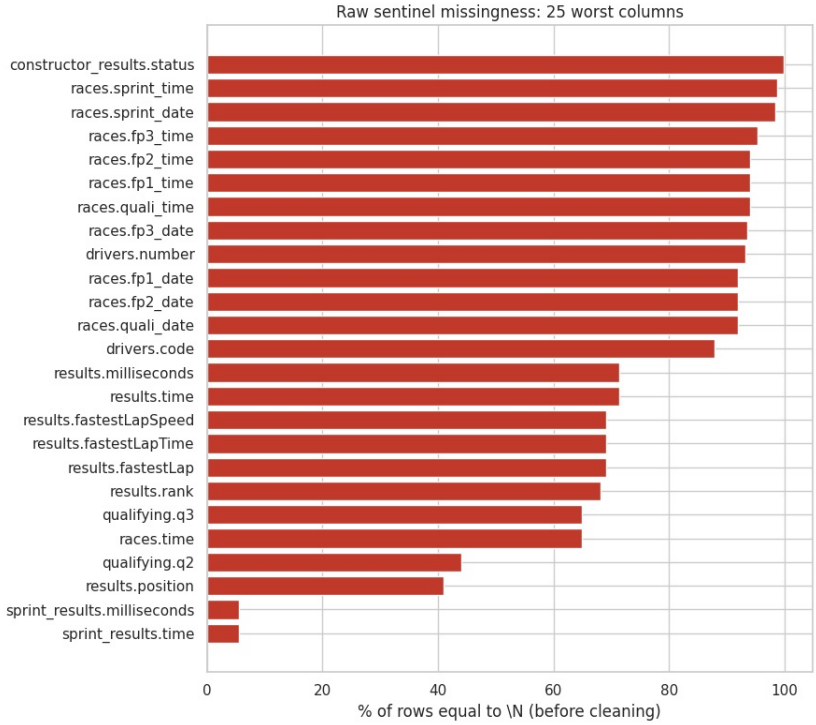


Figure 1. Raw sentinel rate for the 25 worst columns, before cleaning.

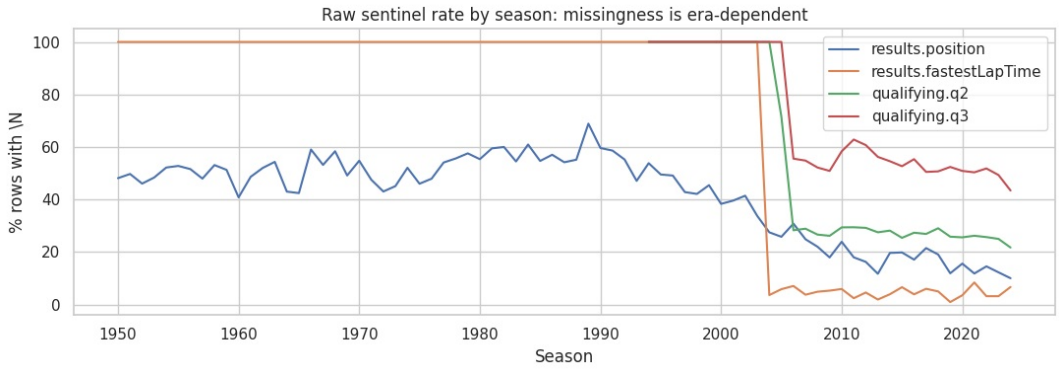


Figure 2. Raw sentinel rate by season for selected columns. Qualifying Q2/Q3 and fastest-lap fields are missing in whole eras, not at random.

Implementation choices. (1) `\N` and empty strings become real missing values. (2) A column is converted to numeric

only if every non-missing value parses, which protects mixed columns such as `positionText` (contains R, D, W). (3) Dates (`races.date`, practice/qualifying/sprint dates, `drivers.dob`) become datetimes. (4) Lap-time strings like `1:27.452` become numeric seconds. The histograms of the converted durations were checked for parsing outliers.

Check	Before cleaning	After cleaning
Sentinel (\N) cells	160,076	0
Text / numeric / datetime columns	120 / 0 / 0	33 / 80 / 7
Result rows sharing a (raceId, driverId)	91 extra rows	0
Result rows	26,759	26,668
Orphan foreign-key values	0	0

1.2 Keys

Primary keys were checked for uniqueness and non-null values in all 14 tables (composite keys for `lap_times` and `pit_stops`), and 23 foreign-key relations were reconciled against their parent tables. **No orphans were found**, so joins do not silently drop or invent rows.

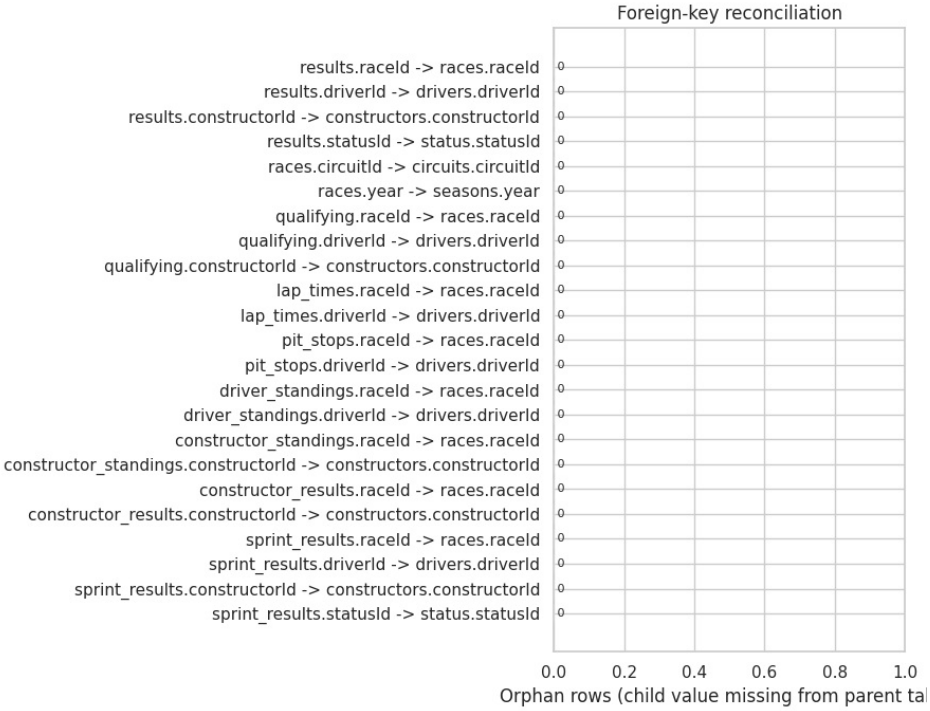


Figure 3. Foreign-key reconciliation: orphan rows per relation (all zero).

1.3 Grain: exactly one row per (raceId, driverId)

The results table has its own key (`resultId`), but a prediction is one driver in one race, so `(raceId, driverId)` must be unique. We found **85 pairs with more than one row (176 rows, 91 extra)**, mostly shared drives in 1950–1964, plus a few 1978 entries (14 seasons in total).

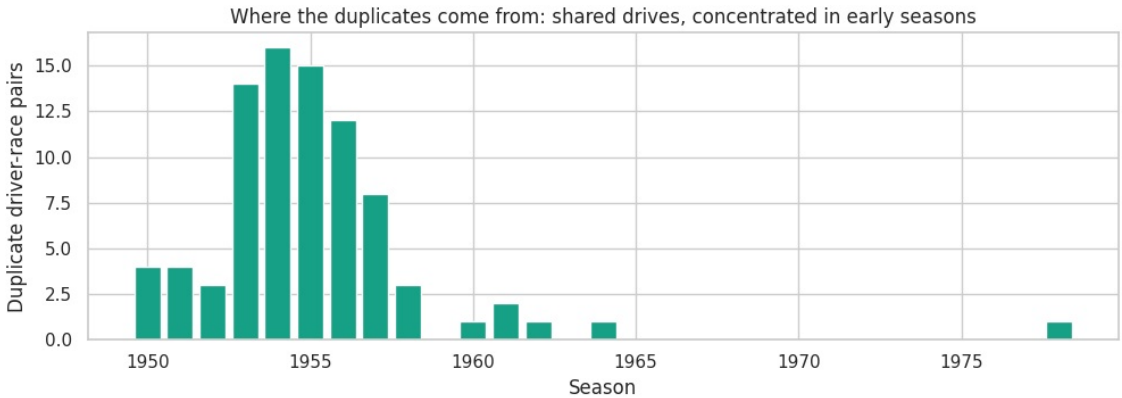


Figure 4. Duplicate driver-race pairs by season: shared drives, concentrated in the early years.

> **Rule chosen:** keep the row with the best (lowest non-zero) `grid` position, break ties by the lowest `resultId`. **Why:** the rule uses only information that exists before the race. The obvious alternative, keeping the best finishing position, chooses rows using the label and would inflate podium rates for shared-drive drivers. **Sensitivity:** under the alternative rule only **17 of 26,668 labels** change, and the podium rate moves from 12.67% to 12.73%, so the choice has negligible effect on the results.

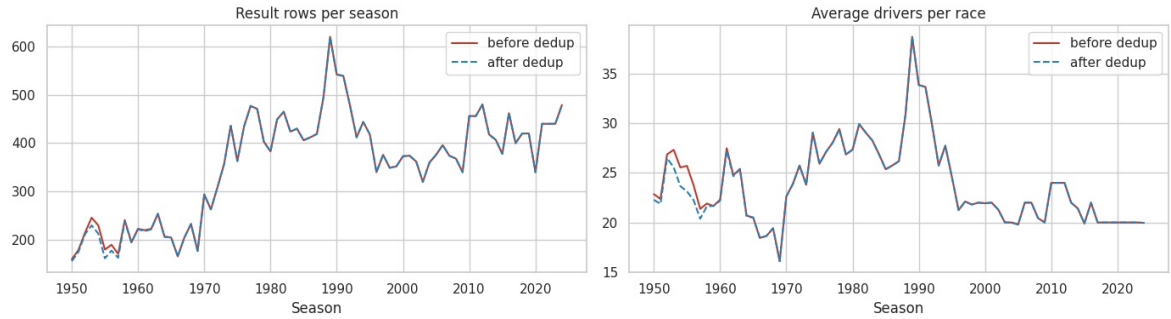


Figure 5. Effect of the grain fix on rows per season and average field size (91 rows removed).

2. Exploratory analysis

After cleaning, the modeling table has 26,668 rows and a podium rate of **12.67%** (3,378 podiums). The positive class is rare, so accuracy is misleading (predicting “no podium” for everyone scores about 87%) and the project uses PR-AUC as its primary metric. Starting position is the dominant pre-race signal: about 63% of pole-sitters, 55% of P2 and 46% of P3 starters finish on the podium, falling steeply beyond the fifth row. The podium rate per season moves between roughly 8% and 19%, so the era matters. Qualifying records exist only from **1994** and cover only part of the races until 2003 (5–60% of races in 1994–2002, 100% afterwards); Q2 and Q3 times exist only from 2006. Before that, grid position is the only pre-race signal (limitation, section 6).

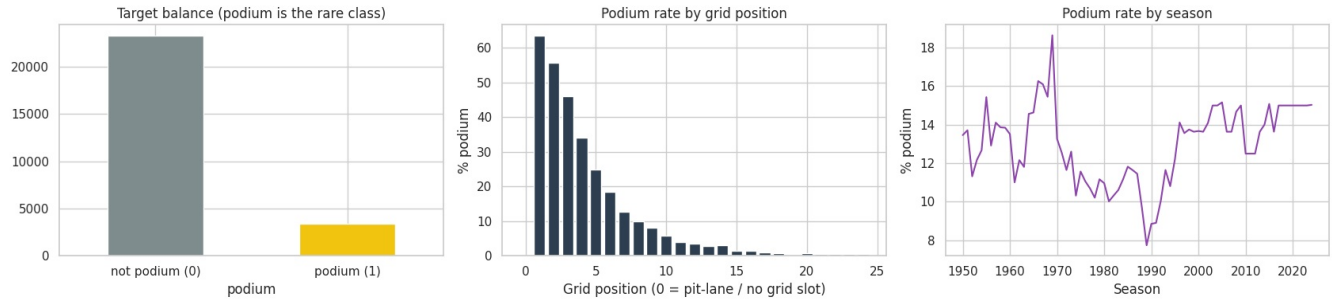


Figure 6. Target balance, podium rate by grid position, and podium rate by season.

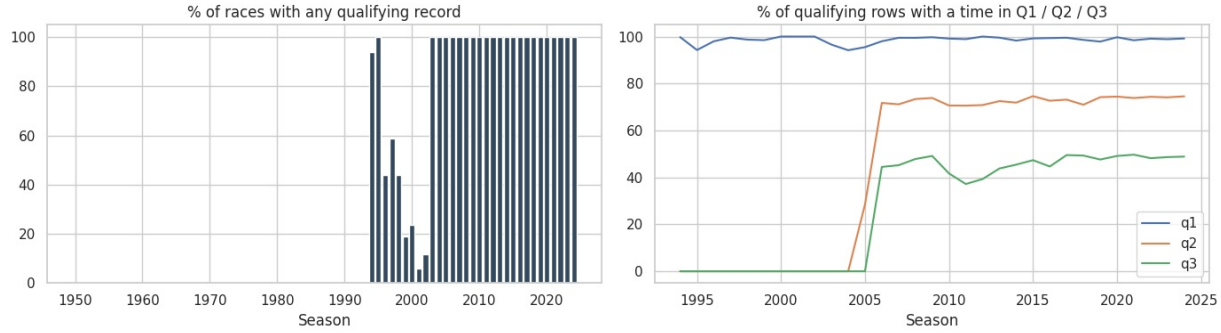


Figure 7. Qualifying coverage drift: share of races with a qualifying record and share of rows with a Q1/Q2/Q3 time, by season.

3. Data-engineering questions

Grain for all three: one driver in one race (after deduplication). Joins use `results`, `qualifying`, `races`, `circuits`, `drivers`, `constructors` and `status` with many-to-one validation to protect the grain.

Q1. Which circuits convert a front-row start into a podium most often, and does the ranking change after 2014?

Method. Qualifying position and grid position are kept separate: qualifying comes from `qualifying`, the grid from `results` (after penalties). Qualifying exists from 1994, so the analysis covers 10,553 rows (1994–2024); 0.6% of rows in covered races have no qualifying position. Front-row starts: 988 by qualifying, 986 by grid. Circuits need at least 30 front-row starts

(17 of 43 qualify), and rates carry 95% Wilson intervals. The pre-race cutoff is respected: both positions are known before the start.

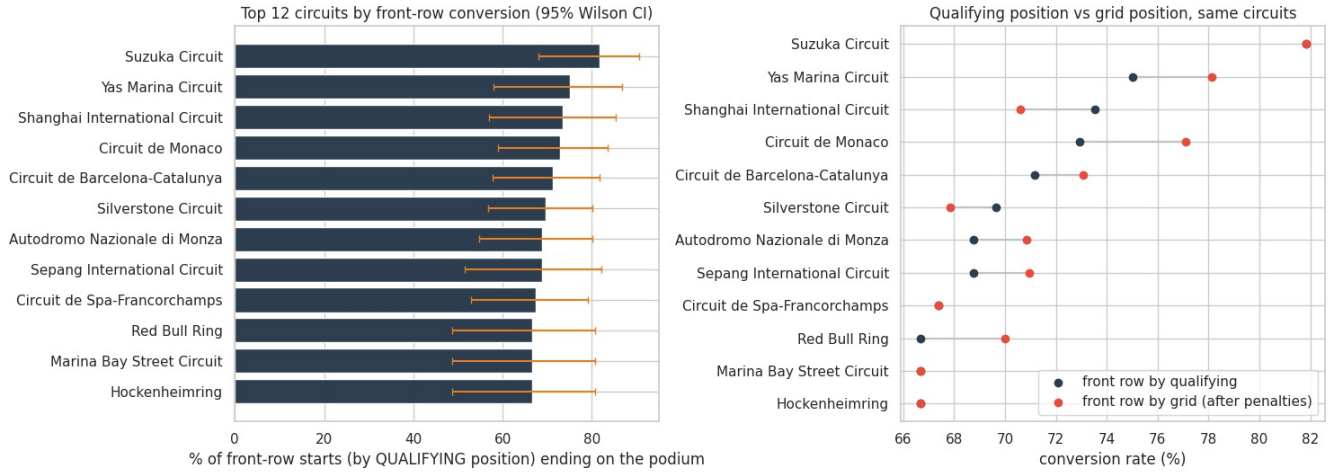


Figure 8. Left: top circuits by front-row conversion (qualifying definition) with 95% intervals. Right: the same circuits when the front row is defined by the grid instead.

> **Answer.** Suzuka converts front-row starts best (81.8% of 44), followed by Yas Marina (75.0% of 32) and Shanghai (73.5% of 34). The two definitions differ because **3.8% of front-row qualifiers did not start on the front row**: only 28.9% of those 38 drivers finished on the podium, against 70.4% of those who did, so penalties matter most at circuits where they are common.

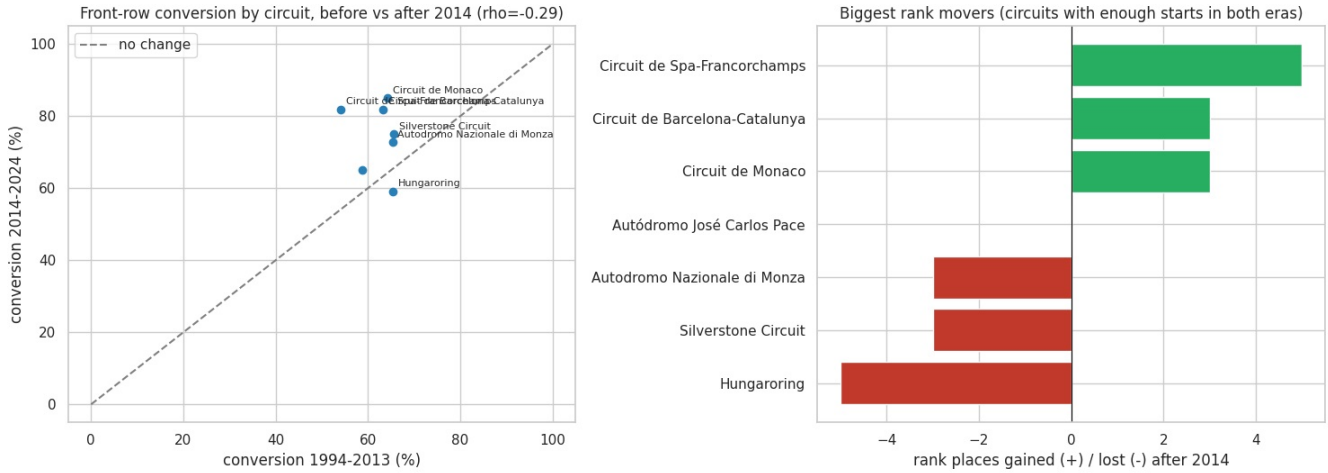


Figure 9. Before vs after the 2014 hybrid era (circuits with at least 20 front-row starts in both eras).

> **Did the ranking change after 2014?** Overall front-row conversion rose from 64.7% (1994–2013) to 73.7% (2014–2024). Only **7 circuits** have enough starts in both eras, and their rankings correlate at Spearman ρ = −0.29 (p = 0.53) by the qualifying definition and 0.00 by the grid definition. The biggest climbers are Spa (+5 places), Barcelona (+3) and Monaco (+3); the biggest fallers are the Hungaroring (−5), Silverstone (−3) and Monza (−3). With only 7 circuits and wide intervals this is not statistically significant, so the honest conclusion is that the order is **not stable** and shows no reliable agreement across eras, while the overall level rose.

Q2. Do drivers podium more often at a circuit in their home country, after controlling for starting position?

Method. The driver's nationality is mapped to a country and compared with the circuit's country (`drivers` → `results` → `races` → `circuits`). 18 nationalities have no mapping (2,489 rows) and are treated as “not home”, which is documented as a limitation. Rows with grid 0 (no defined start position) are dropped, leaving 25,033 rows with 2,317 home starts (9.3%). Home flag and grid are both known before the race. We (i) compare home and away podium rates within grid bands and (ii) fit logistic regressions with standard errors clustered by driver.

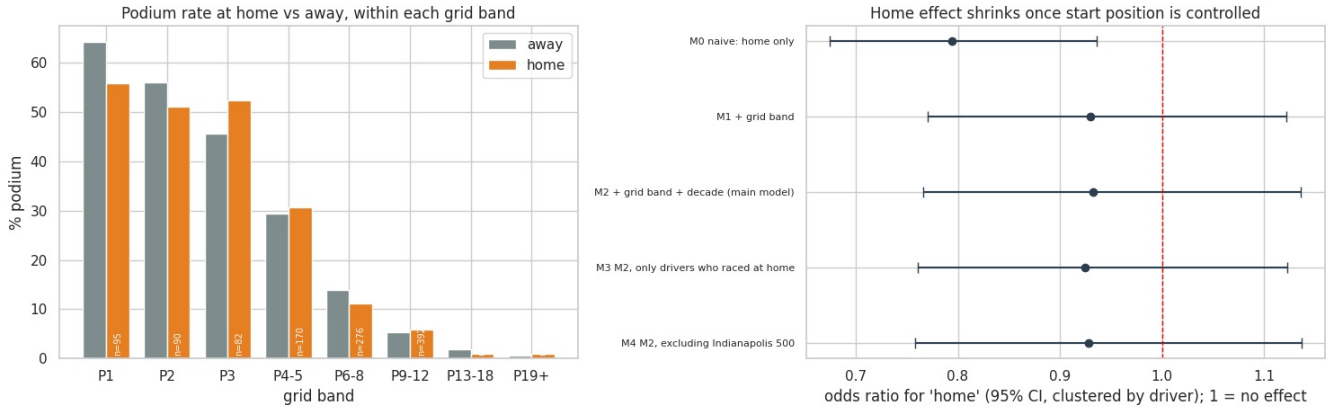


Figure 10. Left: podium rate at home vs away inside each grid band. Right: odds ratio for “home” before and after controlling for starting position, with robustness variants.

Model	Odds ratio (home)	95% CI	p
M0 naive, home only	0.79	0.68–0.94	0.006
M1 + grid band	0.93	0.77–1.12	0.448
M2 + grid band + decade (main)	0.93	0.77–1.14	0.488
M3 only drivers who raced at home	0.92	0.76–1.12	0.428
M4 excluding the Indianapolis 500	0.93	0.76–1.14	0.473

Answer. No. Without a control, home starts look *worse* (odds ratio 0.79). Once starting position is controlled, most of that gap disappears, which is consistent with home drivers starting further back on average rather than racing worse. With starting position controlled, the odds ratio is 0.93 and the interval (0.77–1.14) includes 1: **there is no detectable home advantage**, and the result is the same across all robustness checks. This is an association, not a causal claim.

Q3. Which constructors have the highest mechanical-retirement rate in 2014–2021 vs 2022–2024?

Method. Retirements come from joining `results` → `status` → `constructors`. Each of the 66 distinct finishing statuses that occur in 2014–2024 was classified explicitly as finished, non-start, mechanical (core component failures), ambiguous (tyre, wing, unspecified “Retired”) or non-mechanical (crash, driver error, regulation decisions); the figure shows the classes. The **strict** definition counts only core mechanical failures; a **broad** definition adds the ambiguous group as a sensitivity check. The denominator is race *starts* (non-starts removed), and renamed teams are merged into franchises (e.g. Toro Rosso / AlphaTauri / RB). Constructors with too few starts in a period (Caterham, Manor, Marussia) are excluded. Status is a post-race field, so this question is descriptive and is not used as a model feature.

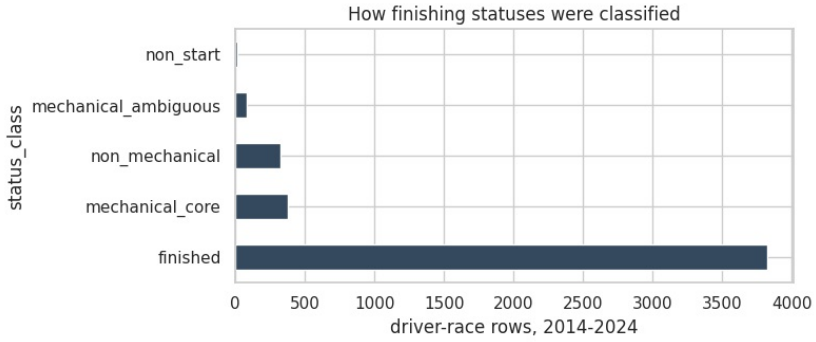


Figure 11. How finishing statuses were classified (2014–2024).

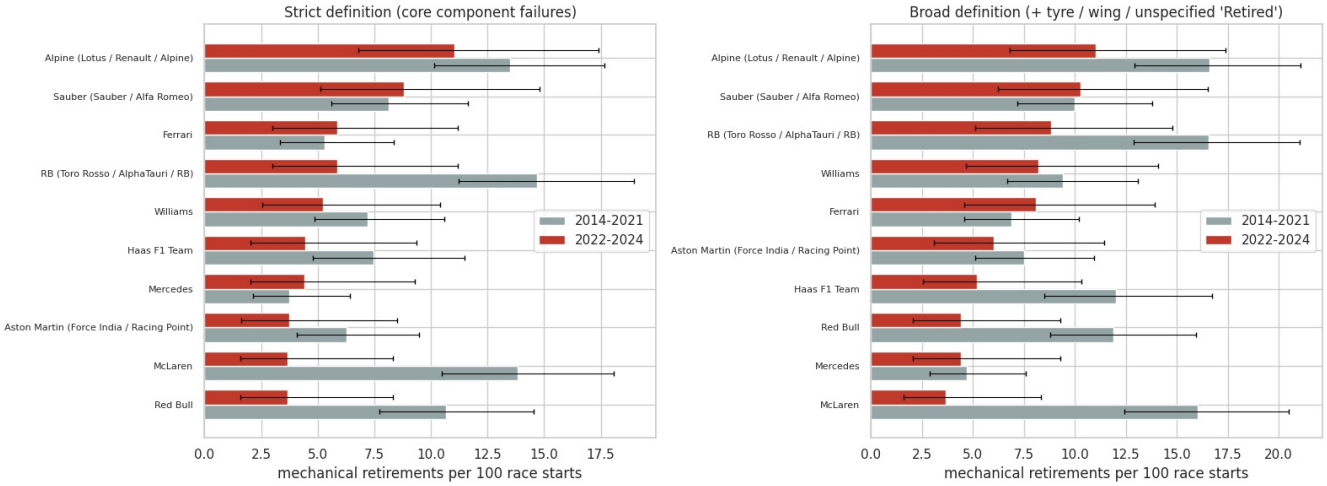


Figure 12. Mechanical retirements per 100 starts by franchise, strict and broad definitions, with 95% intervals.

Period	Highest strict mechanical-retirement rate
2014–2021	RB (Toro Rosso / AlphaTauri / RB) 14.7% (47 of 320) · McLaren 13.8% (44 of 318) · Alpine (Lotus / Renault / Alpine) 13.5% (43 of 319)
2022–2024	Alpine 11.0% (15 of 136) · Sauber (Sauber / Alfa Romeo) 8.8% (12 of 136) · RB 5.9% (8 of 136)

> **Answer.** The leaders changed: RB, McLaren and Alpine in 2014–2021, then Alpine, Sauber and RB in 2022–2024. Fleet-wide, the strict rate fell from 9.2% to 5.7%, and the rank agreement between periods is almost zero (Spearman 0.09). The largest improvements are McLaren (−10.2 percentage points, Fisher p = 0.001) and RB (−8.8 points, p = 0.007); Sauber is essentially unchanged (+0.7 points, p = 0.85). Rates in 2022–2024 rest on only 136 starts per team (three seasons), so most intervals overlap and the rankings of the middle teams should not be over-interpreted.

4. Features, validity at the cutoff, and split

The prediction cutoff is **after qualifying, before the race**. Each feature below was chosen from the EDA and tested by experiment (section 5). Every feature is computed from information that already existed at the cutoff. The final table has 22 features in 6 groups.

Group	Features	Why it is valid before the race
Grid	grid_num, grid_pit	Grid is fixed after qualifying and penalties. A pit-lane start (grid 0) is moved to last place + 1 and flagged.
Qualifying	q_pos, q_gap_pct, has_quali, grid_vs_quali	Qualifying is finished before the race; q_gap_pct is the gap to pole in percent; grid_vs_quali captures penalties; has_quali marks the era without qualifying data.
Driver form	drv_podium_rate, drv_finish_frac, drv_starts_before, drv_podiums_career	Rolling statistics over the driver's **previous** races only (values are shifted by one race before the window is applied).
Team form	con_podium_rate, con_finish_frac	Same idea for the constructor, over its previous races.
Standings	ds_points_frac, ds_pos, ds_wins, cs_points_frac, cs_pos, cs_wins	Standings after the **previous round** of the same season (the table stores standings after each race, so the current race's row is never used). Points are expressed as a fraction of the leader's, which removes the effect of changing points systems.
Context	field_size, round, age, is_home	Entry list, calendar position, date of birth and circuit country are known in advance.

Excluded as leakage: current-race `position`, `positionOrder`, `points`, `laps`, `time`, `milliseconds`, fastest-lap fields, `statusId`, pit stops, lap times and post-race standings. Driver and constructor identifiers are excluded because their hundreds of levels would let a model memorise people instead of learning pre-race signals.

Leakage tests in the notebook: (1) an assertion that no forbidden column is in the feature list; (2) a brute-force recomputation of the driver-form feature for 300 random rows from past rows only; (3) a perturbation test that flips the outcome of one race and verifies that the features of that race and all earlier races are unchanged while later races change.

4.1 Chronological split

Split	Seasons	Races	Rows	Podiums	Podium rate
Train	1950–2016	956	23,289	2,871	12.33%
Validation	2017–2020	79	1,580	237	15.00%
Test	2021–2024	90	1,799	270	15.01%

The split is chronological. A random split would put races of the same weekend and season in both train and test and inflate scores; in real use we predict the future from the past. The validation block lies between train and test so hyperparameters and thresholds are tuned on data older than the test. The test block includes the 2022 regulation change, which makes it an honest test of drift. Validation and test have a higher podium rate (15%) than training (12.3%) because modern fields are smaller than in the 1970–1990s, so PR-AUC values are not directly comparable to the training base rate.

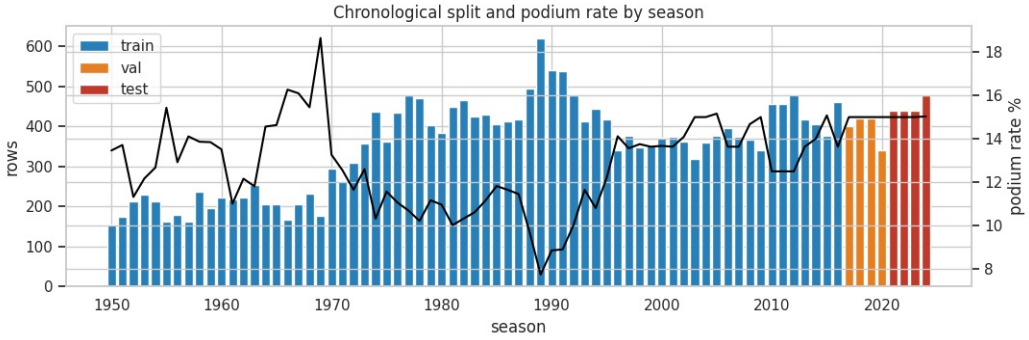


Figure 13. Chronological split: rows and podium rate per block.

4.2 Evidence for the feature choices

Form window. Rolling windows of 3, 5 and 10 previous races were compared on training rows by mean univariate ROC-AUC of driver and team podium rate: 0.790, 0.818 and 0.835. A window of **W = 10 races** was chosen. Longer windows smooth out single bad weekends; the trend was still rising, so W = 10 is the largest value tested, not a proven optimum.

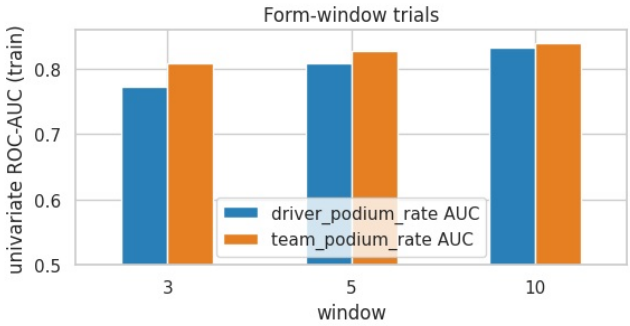


Figure 14. Form-window experiment (mean univariate AUC of driver and team podium rate).

Univariate strength. Each feature was scored by its ROC-AUC on the training rows, shown as strength = max(AUC, 1−AUC). The strongest signals are qualifying position (0.902), grid position (0.888), gap to pole (0.877), driver standing (0.841), constructor podium rate (0.839), constructor standing (0.831) and driver podium rate (0.831). The weakest are `round` (0.503), `is_home` (0.510), `has_quali` (0.519), `grid_pit` (0.538) and `grid_vs_quali` (0.540). Weak univariate features are kept only where the hypothesis justifies them: `is_home` follows Q2 (no effect expected), `grid_pit` and `grid_vs_quali` flag penalties, and `has_quali` marks the era without qualifying data. Their contribution is tested in the ablation (section 5.2).

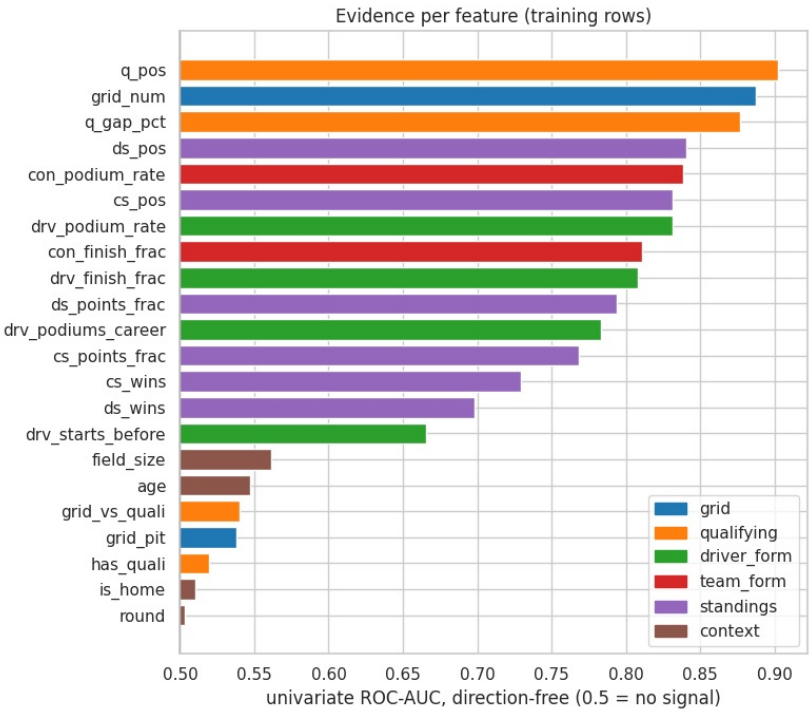


Figure 15. Univariate ROC-AUC strength of each feature on the training rows.

Redundancy. Several features carry overlapping information (Spearman $|\rho| \geq 0.8$): grid and qualifying position (0.98), qualifying position and gap to pole (0.88), grid and gap to pole (0.86), points fraction and position in both championships (-0.91 driver, -0.92 constructor), and constructor podium rate with constructor position (-0.81). This is why single-feature importance must be read with care and why the ablation removes whole groups.

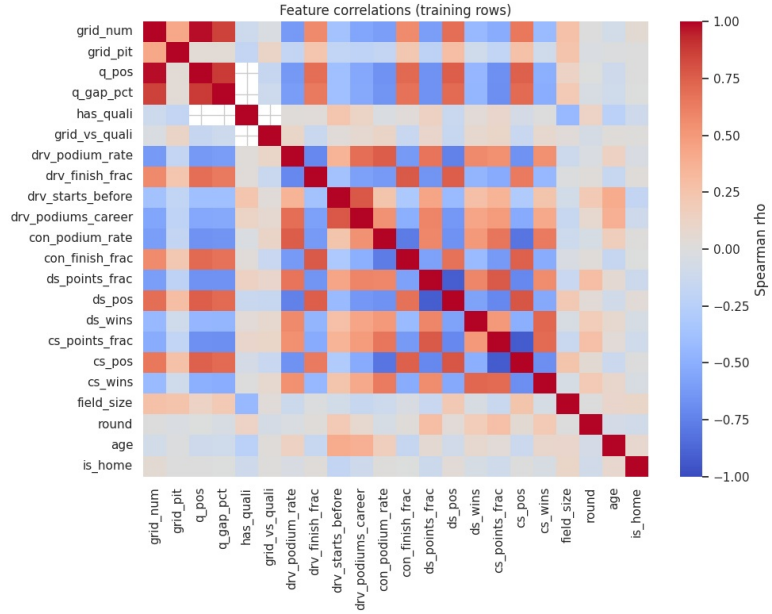


Figure 16. Spearman correlation heatmap of the features.

Preprocessing. Numeric features are median-imputed and standardised, with statistics fitted on the training rows only. The effect on ranges is shown below: for example `q_gap_pct` ranges from 0 to 923 raw (a few extreme gaps from old qualifying sessions), with 16,287 missing values in the training rows, and all inspected features have mean 0 and standard deviation 1 after scaling with no missing values.

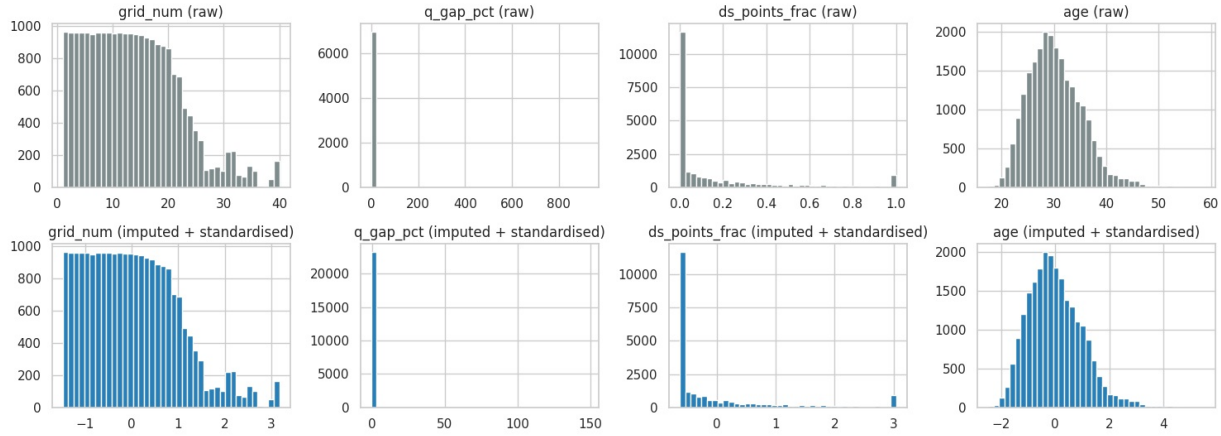


Figure 17. Distribution of key features before and after preprocessing.

5. Models, experiments and explanations

Models (same rows, same features): logistic regression (baseline), gradient boosting, random forest and a shallow neural network (1–2 hidden layers), plus two reference points: the rule “grid ≤ 3” and a grid-only logistic regression. Hyperparameters were chosen on the validation set by PR-AUC, and the decision threshold maximises F1 on the validation set. The test set was used only for the final report. **Primary metric: PR-AUC**, because podiums are rare (about 1 in 8 overall, 1 in 7 in the validation and test blocks) and it measures how well the few true podium finishers are ranked at the top. **Secondary: ROC-AUC**; F1, precision and recall at the validation-chosen threshold are also reported.

5.1 Model comparison

Search grids and chosen configurations (23 trained configurations in total): logistic regression C ∈ {0.01, 0.1, 1, 10} → **C = 10**; gradient boosting learning rate ∈ {0.03, 0.1}, depth ∈ {3, 5}, L2 ∈ {0, 1} → **lr = 0.03, depth 5, L2 = 1**; random forest (3 settings) → **min leaf 20, depth 10**; shallow NN (hidden layers 16, 32, 32×16, 64×32; weight decay 1e-4 or 1e-2) → **one hidden layer of 32 units, alpha = 0.01**. Two-layer networks did worse (validation PR-AUC 0.64–0.69), which suggests extra depth only adds variance on 22 tabular features.

Model	Train PR	Train ROC	Train F1	Val PR	Val ROC	Val F1	Test PR	Test ROC	Test F1	Test Prec	Test Rec
Rule: grid ≤ 3	0.480	0.8878	0.529	0.685	0.9256	0.715	0.646	0.9114	0.661	0.662	0.659
LR (grid only)	0.480	0.8879	0.529	0.685	0.9256	0.715	0.646	0.9114	0.661	0.662	0.659
Logistic Regression	0.562	0.9037	0.559	0.706	0.9361	0.721	**0.721**	0.9354	**0.678**	0.631	0.733
Gradient Boosting	0.642	0.9289	0.581	0.708	0.9424	0.729	0.704	0.9353	0.659	0.683	0.637
Random Forest	0.660	0.9306	0.593	0.720	0.9437	0.716	**0.722**	0.9397	**0.678**	0.678	0.678
Shallow NN	0.602	0.9151	0.553	**0.721**	0.9426	0.724	0.689	0.9304	0.631	0.660	0.604

Thresholds chosen on validation (F1-maximising): rule 0.333, grid-only LR 0.422, LR 0.354, GB 0.430, RF 0.423, NN 0.435. Validation precision/recall: LR 0.641/0.823, GB 0.700/0.760, RF 0.677/0.760, NN 0.702/0.747. Bold = best in column among the four feature models.

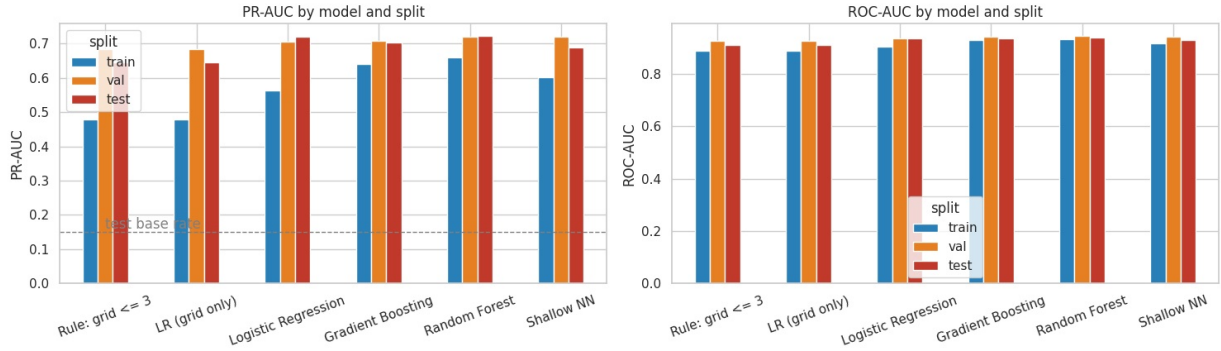


Figure 18. PR-AUC and ROC-AUC by model and split.

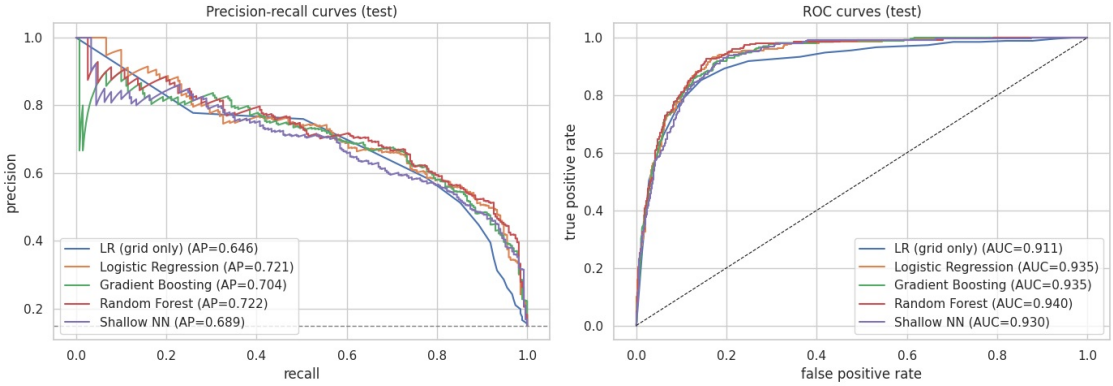


Figure 19. Precision-recall curves on the validation and test sets.

Baseline vs complex models. The grid-only logistic regression is identical to the rule "grid ≤ 3" at its chosen threshold (test PR-AUC 0.646, F1 0.661): a single feature already captures most of the signal. Adding the other 20 features lifts test PR-AUC by +0.04 to +0.08 (to 0.689–0.722) and ROC-AUC from 0.911 to 0.930–0.940, so the extra features help ranking. They help the thresholded decision much less: test F1 ranges from 0.631 to 0.678 against 0.661 for the rule, and the neural network is *below* the rule. The simplest feature model, logistic regression, matches the random forest on test (PR-AUC 0.721 vs 0.722, F1 0.678 for both) and beats gradient boosting and the neural network, so added model complexity did not earn its cost here.

Train vs validation vs test. Training scores are *lower* than validation and test for every model (e.g. random forest PR-AUC 0.660 vs 0.720), the opposite of classic overfitting. The reason is the data, not the model: the training block contains 67 seasons including early eras with long grids, unreliable cars and no qualifying data, where podiums are harder to rank, while 2017–2024 has 20-car fields and full qualifying. The real overfitting signal is the validation-to-test change: the neural network loses 0.033 PR-AUC (0.721 → 0.689) and 0.143 recall (0.747 → 0.604), and gradient boosting loses 0.005 PR-AUC, while logistic regression and random forest are stable or slightly better. The validation block has only 237 podiums, so differences of 0.01–0.02 PR-AUC between the four models are within noise and should not be over-read.

> **Final model choice.** The selection rule fixed in advance was "best validation PR-AUC", which picks the **shallow neural network** (0.7213, ahead of random forest 0.7197 by 0.0016, which is noise). The test set then shows the network as the weakest of the four feature models. We keep the pre-declared rule for the inference function and explanations, and report this honestly; on the test evidence, logistic regression is the more defensible deployment choice because it is the simplest model and is within 0.001 PR-AUC of the best.

Limitations of the models. Logistic regression is linear in its inputs and cannot capture interactions such as "good grid slot in a weak car" unless features encode them; the tree ensembles handle interactions but are sensitive to the era mix; the small network is a black box, trained on the same 22 features without any extra signal such as weather, tyre strategy or reliability, which explains why all models plateau around 0.70–0.72 PR-AUC. Podiums in wet or chaotic races are essentially unpredictable from pre-race data (see the Gasly example in section 5.4).

5.2 Experiments

Feature-set build-up (gradient boosting). Groups are added in the order grid, qualifying, driver form, team form, standings, context.

Feature set	# features	Val PR-AUC	Val ROC-AUC	Val F1	Test PR-AUC	Test ROC-AUC	Test F1
grid	2	0.6852	0.9263	0.715	0.6459	0.9106	0.661
+ qualifying	6	0.7124	0.9323	0.710	0.6544	0.9237	0.668
+ driver form	10	0.7247	0.9417	0.722	0.6960	0.9306	0.656
+ team form	12	0.7272	0.9429	0.728	0.6877	0.9324	0.653
+ standings	18	0.7042	0.9414	0.730	0.6990	0.9342	0.651
+ context	22	0.7084	0.9424	0.729	0.7035	0.9353	0.659

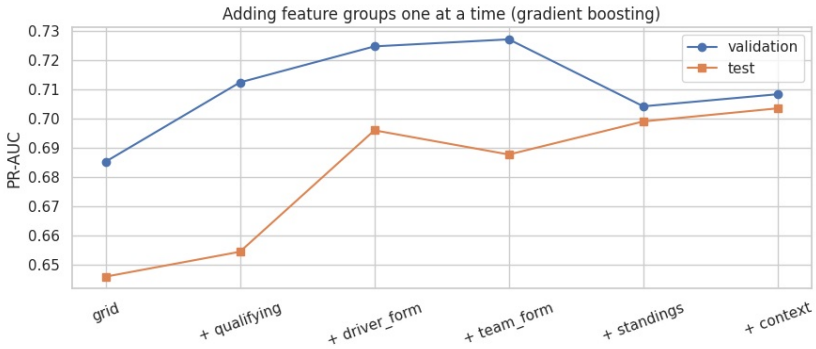


Figure 20. Feature-set build-up: validation and test PR-AUC as groups are added.

Qualifying and driver form give the clearest gains (test PR-AUC 0.646 → 0.696). Standings lower validation PR-AUC by 0.023 but raise test PR-AUC slightly, so the build-up supports keeping them only weakly.

Leave-one-group-out ablation. Each group is removed in turn, the model is retrained, and the change in PR-AUC versus the full model is recorded (negative = the group was helping).

Removed group	GB Δ val	GB Δ test	LR Δ val	LR Δ test
grid	−0.0016	−0.0003	−0.0125	−0.0183
qualifying	−0.0172	+0.0071	+0.0021	−0.0070
driver form	+0.0010	+0.0132	+0.0029	+0.0052
team form	−0.0110	+0.0062	−0.0004	−0.0037
standings	+0.0228	−0.0013	+0.0003	−0.0060
context	−0.0041	−0.0045	−0.0016	−0.0029

Full-model PR-AUC: gradient boosting 0.7084 val / 0.7035 test; logistic regression 0.7061 val / 0.7210 test.



Figure 21. Leave-one-group-out ablation for gradient boosting and logistic regression (change in PR-AUC).

Reading the ablation. Only one result is consistent across both models and both blocks: removing the grid group hurts logistic regression (−0.013 val, −0.018 test), the largest drop in the table. For gradient boosting, validation and test often disagree in sign (qualifying, driver form, standings), and several removals *improve* test PR-AUC. The explanation is redundancy: grid and qualifying position correlate at 0.98 and form and standings measure the same underlying car and driver quality (figure 16), so the remaining groups cover for the removed one. We conclude that grid position is the one indispensable group and that the others add small, partly interchangeable gains. Group removals of ±0.01 are within the noise of this evaluation and should not be used to claim that a group is useless.

Preprocessing-order experiments (logistic regression).

Experiment	Val PR-AUC	Val ROC-AUC	Test PR-AUC	Test ROC-AUC
A1 impute → scale (train statistics)	0.7061	0.9361	0.7210	0.9354
A2 scale → impute (train statistics)	0.7063	0.9360	0.7208	0.9355
B1 fit on TRAIN only (correct)	0.7061	0.9361	0.7210	0.9354
B2 fit on ALL rows (leaky)	0.7064	0.9361	0.7208	0.9353
C1 winsorise → scale	0.7049	0.9365	0.7236	0.9358
C2 scale → clip at ±3 sd	0.7063	0.9368	0.7226	0.9353

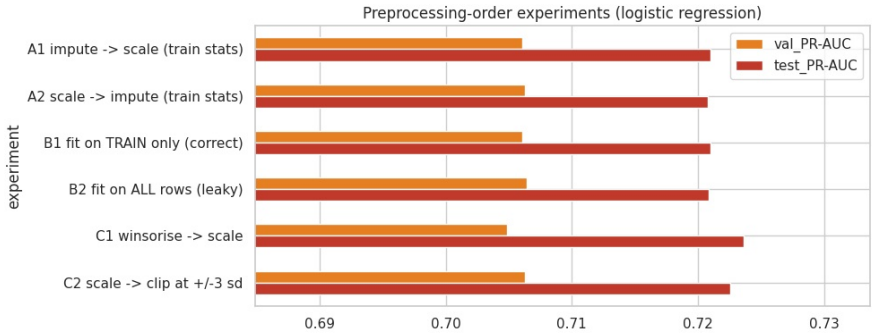


Figure 22. Preprocessing-order experiments: validation and test PR-AUC.

All orderings land within 0.0014 validation PR-AUC (A: +0.0002, B: +0.0003, C: +0.0014), so the order does not matter for this data. Median imputation and standardisation commute almost exactly when few values are missing, and the leaky variant (B2) gains only 0.0003 because the shifts in mean and variance are small. The principle still holds: statistics from validation or test rows are information the model could not have had, so the final pipeline fits every step on the training rows only.

Training-window experiment (gradient boosting). Does old data help or hurt the modern test period?

Train from	Train rows	Val PR-AUC	Val ROC-AUC	Test PR-AUC	Test ROC-AUC
1950	23,289	0.7084	0.9424	0.7035	0.9353
1980	15,453	0.7029	0.9423	0.6971	0.9355
1994	8,979	0.7047	0.9421	0.7020	0.9313
2006	4,535	0.7281	0.9432	0.6739	0.9274

Using all seasons (1950 onward) is as good as any shorter window on test, so there is no evidence that early history hurts and we keep it. Training only from 2006 gives the best validation PR-AUC (0.728) but the worst test PR-AUC (0.674), a reminder that a four-season validation block can mislead the selection.

5.3 Global explainability

The explained model is the final one (shallow neural network). Permutation importance is computed on the validation set (drop in PR-AUC when a feature is shuffled, repeated, with error bars); SHAP uses a permutation explainer on a sample of rows.

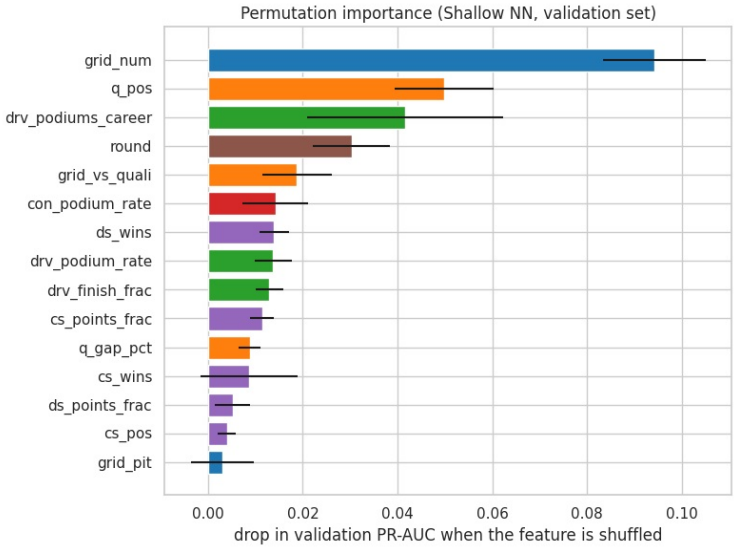


Figure 23. Permutation importance (shallow NN, validation set).

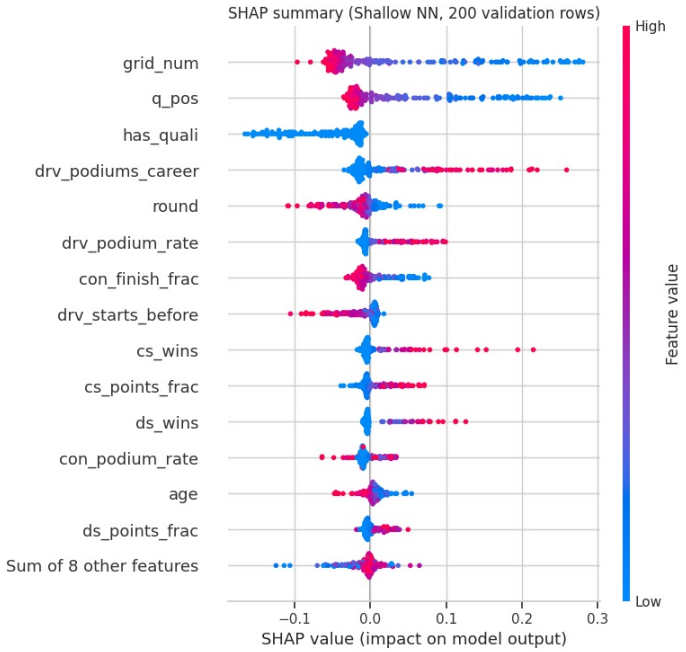


Figure 24. SHAP summary (beeswarm) plot.

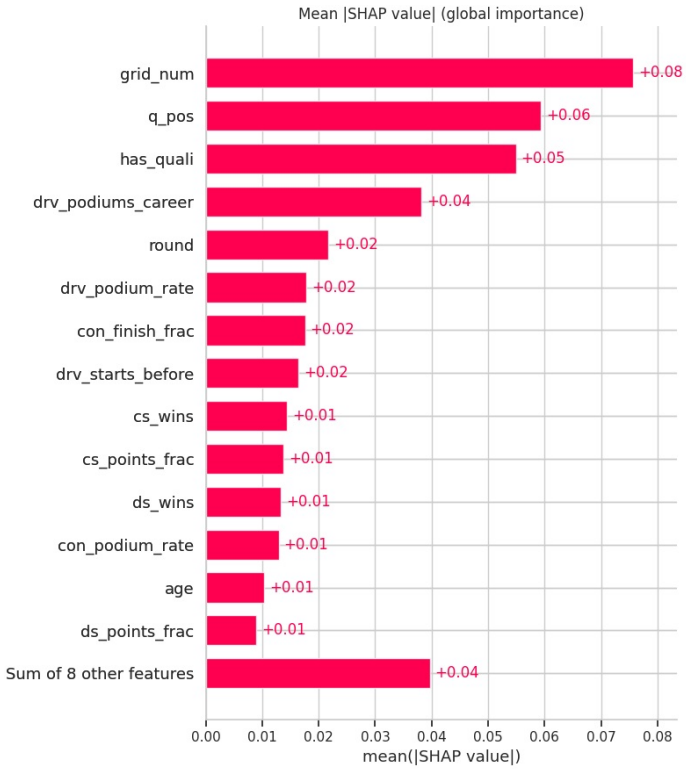


Figure 25. SHAP global importance (mean |SHAP|).

Rank	Permutation importance (drop in val PR-AUC)	Mean |SHAP|
1	grid_num 0.094	grid_num 0.076
2	q_pos 0.050	q_pos 0.059
3	drv_podiums_career 0.042	has_quali 0.055
4	round 0.030	drv_podiums_career 0.038
5	grid_vs_quali 0.019	round 0.022
6	con_podium_rate 0.014	drv_podium_rate 0.018
7	ds_wins 0.014	con_finish_frac 0.018
8	drv_podium_rate 0.014	drv_starts_before 0.016

Interpretation. Both methods agree on the top four: the model mostly relies on where the driver starts (`grid_num`, `q_pos`) and on the driver's career podium count, with the calendar position `round` also important. This matches the EDA (figure 6) and the ablation: grid is the dominant signal. Two results are surprising. `round` has almost no univariate power (AUC 0.503) yet ranks fourth, and `has_quali` ranks third in SHAP but is ignored by permutation importance. Our hypothesis, which we did not test, is that these features act as era markers (modern seasons have more rounds and always have

qualifying data) that let the network adjust the base podium rate by era, rather than describing racing. Team-form and standings features have small individual importance because they overlap with each other and with grid.

5.4 Local explainability

Two test-set predictions were explained with SHAP waterfall plots and LIME: the podium the model was most sure of, and a podium it missed.

- **Example 1 — Verstappen, Abu Dhabi GP 2023 (Red Bull), grid 1, P(podium) = 0.98.** LIME attributes the largest positive weights to `grid_num ≤ 7` (+0.187), `drv_podiums_career > 11` (+0.094), `q_pos ≤ 11` (+0.062), `q_gap_pct ≤ 1.91` (+0.053) and `con_podium_rate > 0.20` (+0.045). The prediction is explained by a front-row start in a successful car.
- **Example 2 — Gasly, São Paulo GP 2024 (Alpine), grid 13, P(podium) = 0.00.** He finished on the podium, but every pre-race signal points the other way: `q_pos > 11` (−0.057), `q_gap_pct > 1.91` (−0.056), `con_podium_rate ≤ 0` (−0.027) and `cs_wins ≤ 0` (−0.018). The model is not wrong given its inputs; the result came from race-day events (wet, interrupted race) that no pre-race feature can contain.

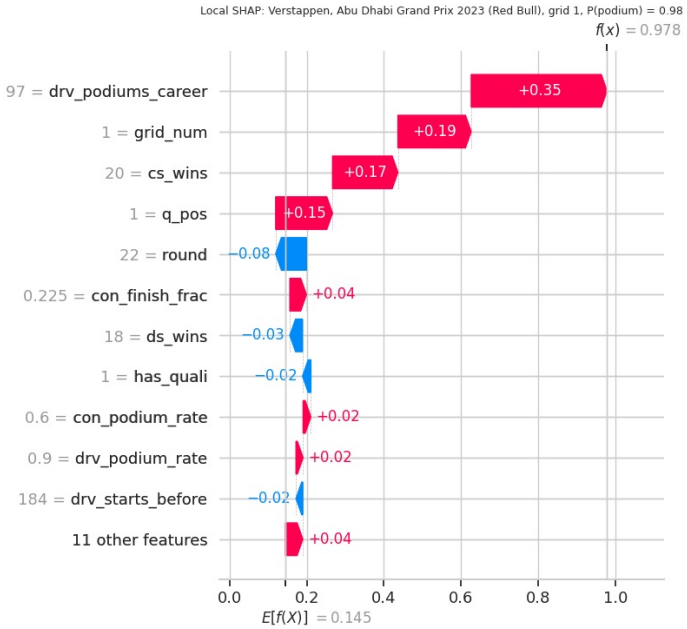


Figure 26. SHAP waterfall, Example 1 (Verstappen, Abu Dhabi 2023).

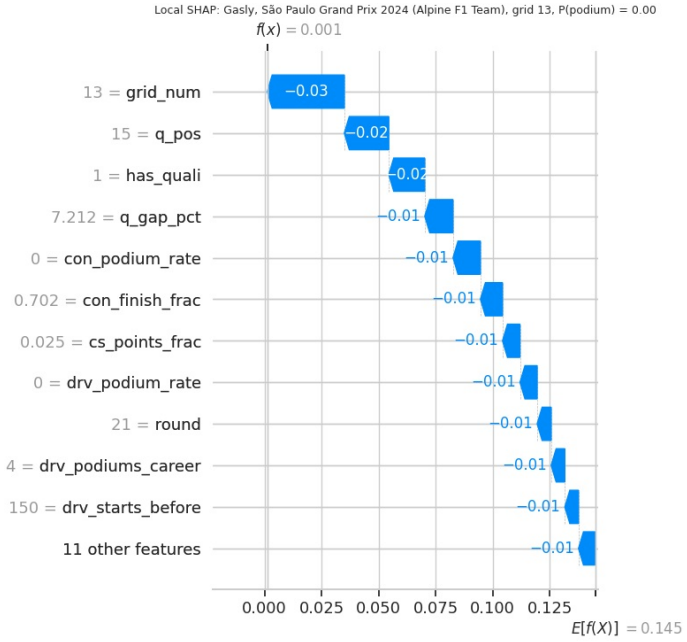


Figure 27. SHAP waterfall, Example 2 (Gasly, São Paulo 2024).

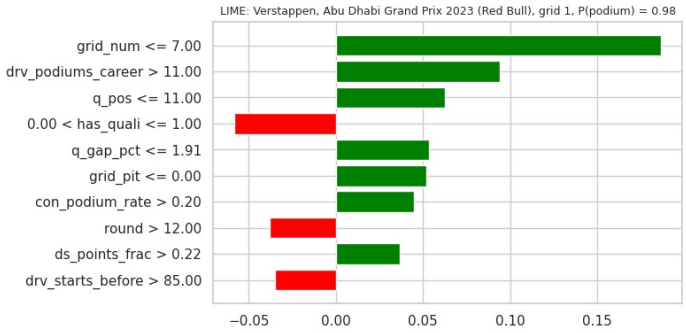


Figure 28. LIME explanation, Example 1.

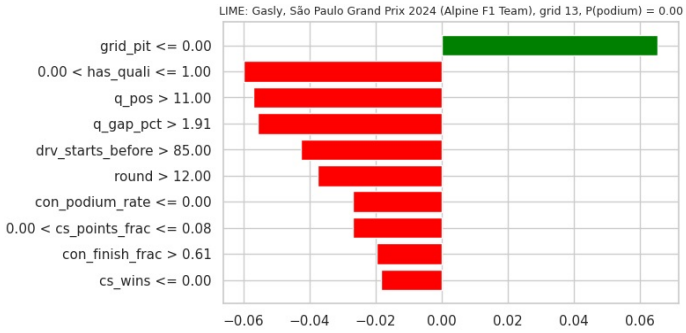


Figure 29. LIME explanation, Example 2.

Important: permutation importance, SHAP and LIME describe what the *model relies on* to make predictions. They are **not causal effects**: a feature can matter because it is correlated with something real (a fast car), and correlated features (grid and qualifying position; driver form and standings) share credit, so low importance for one of them does not mean it is irrelevant. LIME fits a local linear model and can vary between runs, so it is used as a second opinion beside SHAP.

5.5 Inference function

`predict_podium(record)` takes one driver-race record as a dictionary, ignores unknown fields, accepts missing features (filled by the median imputer fitted on the training data), validates impossible values (the third demo row passes an invalid grid value), and returns the podium probability, the thresholded decision and a list of notes. The model and threshold (0.435) are saved as a bundle (`f1_podium_model.joblib`).

Record	P(podium)	Predicted podium	Missing features	Notes
Complete record (real outcome: podium)	0.9782	True	0	—
Partially missing record	0.1913	False	16 of 22	unknown field `driver_name` ignored
Invalid grid value	0.0144	False	17 of 22	unknown field ignored; invalid grid value flagged

With 16 of 22 features missing (all qualifying, form and standings fields), the probability falls from 0.98 to 0.19, so the function degrades toward the median driver instead of failing. This is a sensible fallback but not a calibrated one: a record with this little information should be treated as low-confidence.

6. Known limitations

- **Qualifying-coverage drift.** Qualifying records start in 1994 and cover only part of the races until 2003; Q2/Q3 times exist only from 2006. About 61% of training-table rows have no qualifying features, and models lean on grid position and form for those races. Q1 covers 1994–2024 only.
- **Era sensitivity.** Points systems, reliability, field sizes and regulations changed over 75 seasons, and the podium base rate varies between roughly 8% and 19% by season. Standings are normalised by the leader's points, but drift remains; the 2022 regulation change lies inside the test block. The train rate (12.3%) differs from the validation and test rate (15%).
- **Small samples.** Q1 compares only 7 circuits across eras, and Q3 has 136 starts per team in 2022–2024. The validation and test blocks hold 237 and 270 podiums, so model differences of 0.01–0.02 PR-AUC and ablation changes of about ±0.01 are within the uncertainty.
- **Modeling choices, not facts.** The shared-drive rule (17 of 26,668 labels would change), the nationality-to-country mapping used for the home flag (18 unmapped nationalities) and the mechanical-retirement status classification are judgments. Sensitivity checks are reported for each.
- **Single split and model selection.** One fixed chronological split is used. The validation-to-test change was up to

0.033 PR-AUC (shallow NN), and the model picked by validation (the network) is not the best on test, so a different split could reorder the models. No confidence intervals on the metrics were computed; bootstrapping over races would be a natural next step.

- **Unexplained feature behaviour.** The high importance of `round` and `has_quali` is hypothesised to be an era effect but was not tested directly.
- **No race-day information.** Weather, tyre strategy and first-lap incidents are not available at the cutoff, which caps achievable performance (example 2, Gasly).
- **Associations, not causes.** All explanations and Q2 describe statistical associations.

Sources: Ergast Formula 1 World Championship dataset (Kaggle, 1950–2024). All numbers in this report are taken from the executed notebook.